

HAT-P-36b

R-1548

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_200524 · created 2026-08-01T19:45:20+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458993.7221 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.148 +/- 0.015
Transit depth (Rp/Rs)^2	2.18 +/- 0.44 [%]
Orbital Inclination (inc)	82.06 +/- 2.67
Airmass coefficient 1 (a1)	1.066 +/- 0.017
Airmass coefficient 2 (a2)	-0.054 +/- 0.024
Scatter in the residuals of the lightcurve fit is	1.55 %
Best Comparison Star	#1 - [179, 186]
Optimal Aperture	4.06
Optimal Annulus	6.61
Transit Duration (day)	0.082 +/- 0.014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.148 $\pm$ 0.015 vs 0.126 (deviation 1.27 σ)
Duration fit vs published	0.082 d vs 0.0925 d (deviation 0.65 σ)
Mid-time O–C	4.9 min
Depth SNR	8.5
Residual scatter	1.55 %

Light curve

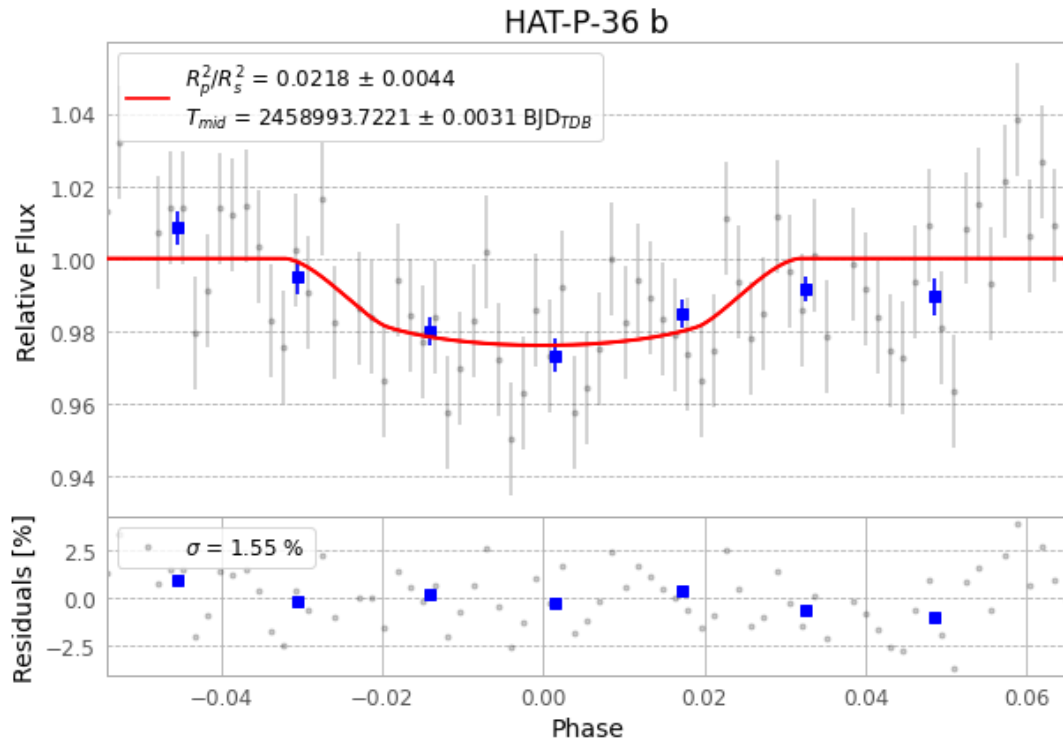


Figure 1 — FinalLightCurve_HAT-P-36 b_2020-05-23.png (SHA-256 762ddb37100b62ae...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [450, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d5f5730a68faf031...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

75 FITS frames (49 MB), HATP-36200524033318.FITS → HATP-36200524072112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200524.log (1e4acd6baf0f...), FinalLightCurve_HAT-P-36 b_2020-05-23.png (762ddb37100b...), FinalLightCurve_HAT-P-36 b_2020-05-23.csv (5f2bf325275b...)

Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 19:45Z.